Login Form :

MainActivity.java

```
package com.example.loginform;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      TextView username = (TextView)findViewById(R.id.username);
      TextView password = (TextView)findViewById(R.id.password);
      Button loginbtn = (Button)findViewById(R.id.loginbtn);
      loginbtn.setOnClickListener(new View.OnClickListener() {
         @Override
         public void onClick(View view) {
            if(username.getText().toString().equals("admin") && password.getText().toString().equals("123456"))
            {
               Toast.makeText(MainActivity.this,"Login Successfully:)",Toast.LENGTH_SHORT).show();
            }
            else
            {
               Toast.makeText(MainActivity.this, "Login Failed:(", Toast.LENGTH_SHORT).show();
            }
         }
      });
   }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:background="#8893f7"
   tools:context=".MainActivity">

   <TextView
      android:id="@+id/signin"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:layout_marginStart="50dp"
      android:layout_marginTop="100dp"
      android:layout_marginEnd="50dp"
      android:layout_marginBottom="50dp"
```

```
    android:layout_gravity="center"
    android:text="sign_in"
    android:textColor="@color/white"
    android:textSize="35sp"
    android:textStyle="bold" />

  <EditText
    android:id="@+id/username"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@+id/signin"
    android:layout_marginStart="20dp"
    android:layout_marginTop="20dp"
    android:layout_marginEnd="20dp"
    android:background="#eeccff"
    android:drawableStart="@drawable/baseline_account_circle_24"
    android:drawablePadding="20dp"
    android:hint="username"
    android:padding="20dp" />

  <EditText
    android:id="@+id/password"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@+id/username"
    android:layout_marginStart="20dp"
    android:layout_marginEnd="20dp"
    android:layout_marginBottom="20dp"
    android:background="#eeccff"
    android:drawableStart="@drawable/sharp_admin_panel_settings_24"
    android:drawablePadding="20dp"
    android:hint="password"
    android:inputType="textPassword"
    android:padding="20dp"
    android:autofillHints="" />

  <com.google.android.material.button.MaterialButton
    android:id="@+id/loginbtn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/password"
    android:layout_centerHorizontal="true"
    android:layout_marginStart="20dp"
    android:text="login" />

</RelativeLayout>
```

Simple Calculator

main_activity.java

```
package com.example.calculator2026;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

public class MainActivity extends AppCompatActivity {
   Button button0, button1, button2, button3, button4, button5, button6, button7, button8, button9, buttondot, buttonadd,
buttonsub, buttondiv, buttonmul, buttonC, buttoneql;
   EditText result;
   float mvalueone, mvaluetwo;
   boolean add, sub, mul, div;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      // Initialize buttons
      button0 = findViewById(R.id.buttonzero);
      button1 = findViewById(R.id.button1);
      button2 = findViewById(R.id.button2);
      button3 = findViewById(R.id.button3);
      button4 = findViewById(R.id.button4);
      button5 = findViewById(R.id.button5);
      button6 = findViewById(R.id.button6);
      button7 = findViewById(R.id.button7);
      button8 = findViewById(R.id.button8);
      button9 = findViewById(R.id.button9);
      buttondot = findViewById(R.id.buttondot);
      buttonadd = findViewById(R.id.buttonadd);
      buttonsub = findViewById(R.id.buttonsub);
      buttonmul = findViewById(R.id.buttonmul);
      buttondiv = findViewById(R.id.buttondiv);
      buttonC = findViewById(R.id.buttonC);
      buttoneql = findViewById(R.id.buttoneql);

      result = findViewById(R.id.edit1); // Make sure your EditText has android:id="@+id/edt1" in XML

      // Set number buttons
      View.OnClickListener numberListener = new View.OnClickListener() {
         @Override
         public void onClick(View v) {
            Button b = (Button) v;
            result.append(b.getText().toString());
         }
      };

      button0.setOnClickListener(numberListener);
      button1.setOnClickListener(numberListener);
      button2.setOnClickListener(numberListener);
      button3.setOnClickListener(numberListener);
```

```
button4.setOnClickListener(numberListener);
button5.setOnClickListener(numberListener);
button6.setOnClickListener(numberListener);
button7.setOnClickListener(numberListener);
button8.setOnClickListener(numberListener);
button9.setOnClickListener(numberListener);
buttondot.setOnClickListener(numberListener);

// Clear button
buttonC.setOnClickListener(new View.OnClickListener() {
  @Override
  public void onClick(View v) {
    result.setText("");
  }
});

// Operation buttons
buttonadd.setOnClickListener(new View.OnClickListener() {
  @Override
  public void onClick(View v) {
    if (!result.getText().toString().isEmpty()) {
      mvalueone = Float.parseFloat(result.getText().toString());
      add = true;
      result.setText("");
    }
  }
});

buttonsub.setOnClickListener(new View.OnClickListener() {
  @Override
  public void onClick(View v) {
    if (!result.getText().toString().isEmpty()) {
      mvalueone = Float.parseFloat(result.getText().toString());
      sub = true;
      result.setText("");
    }
  }
});

buttonmul.setOnClickListener(new View.OnClickListener() {
  @Override
  public void onClick(View v) {
    if (!result.getText().toString().isEmpty()) {
      mvalueone = Float.parseFloat(result.getText().toString());
      mul = true;
      result.setText("");
    }
  }
});

buttondiv.setOnClickListener(new View.OnClickListener() {
  @Override
  public void onClick(View v) {
    if (!result.getText().toString().isEmpty()) {
      mvalueone = Float.parseFloat(result.getText().toString());
      div = true;
      result.setText("");
    }
```

```
        }
    });

    // Equals button
    buttoneql.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            if (!result.getText().toString().isEmpty()) {
                mvaluetwo = Float.parseFloat(result.getText().toString());
                if (add) {
                    result.setText(String.valueOf(mvalueone + mvaluetwo));
                    add = false;
                } else if (sub) {
                    result.setText(String.valueOf(mvalueone - mvaluetwo));
                    sub = false;
                } else if (mul) {
                    result.setText(String.valueOf(mvalueone * mvaluetwo));
                    mul = false;
                } else if (div) {
                    if (mvaluetwo != 0) {
                        result.setText(String.valueOf(mvalueone / mvaluetwo));
                    } else {
                        result.setText("Error");
                    }
                    div = false;
                }
            }
        }
    });
  }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity">

  <EditText
    android:id="@+id/edit1"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter value"
    android:textSize="24sp"
    android:inputType="numberDecimal"
    android:layout_marginBottom="20dp"/>

  <Button android:id="@+id/button1"
    android:layout_width="60dp"
    android:layout_height="60dp"
    android:layout_marginStart="50dp"
```

```
    android:layout_marginTop="150dp"
    android:text="1"
    android:textSize="25dp"
    android:background="#964B00" />

<Button android:id="@+id/button2"
    android:layout_width="60dp"
    android:layout_toEndOf="@id/button1"
    android:layout_height="60dp"
    android:layout_marginStart="10dp"
    android:layout_marginTop="150dp"
    android:text="2"
    android:textSize="25dp"
    android:background="#964B00"/>

<Button android:id="@+id/button3"
    android:layout_width="60dp"
    android:layout_toEndOf="@id/button2"
    android:layout_height="60dp"
    android:layout_marginStart="10dp"
    android:layout_marginTop="150dp"
    android:text="3"
    android:textSize="25dp"
    android:background="#964B00"/>

<Button android:id="@+id/button4"
    android:layout_width="60dp"
    android:layout_below="@id/button1"
    android:layout_marginStart="50dp"
    android:layout_marginTop="10dp"
    android:layout_height="60dp"
    android:text="4"
    android:textSize="25dp"
    android:background="#964B00"/>

<Button android:id="@+id/button5"
    android:layout_width="60dp"
    android:layout_below="@id/button2"
    android:layout_toEndOf="@id/button4"
    android:layout_marginStart="10dp"
    android:layout_marginTop="10dp"
    android:layout_height="60dp"
    android:text="5"
    android:textSize="25dp"
    android:background="#964B00"/>

<Button android:id="@+id/button6"
    android:layout_width="60dp"
    android:layout_below="@id/button3"
    android:layout_toEndOf="@id/button5"
    android:layout_marginStart="10dp"
    android:layout_marginTop="10dp"
    android:layout_height="60dp"
    android:text="6"
    android:textSize="25dp"
    android:background="#964B00"/>

<Button android:id="@+id/button7"
```

```
    android:layout_width="60dp"
    android:layout_below="@id/button4"
    android:layout_marginStart="50dp"
    android:layout_marginTop="10dp"
    android:layout_height="60dp"
    android:text="7"
    android:textSize="25dp"
    android:background="#964B00"/>

<Button android:id="@+id/button8"
    android:layout_width="60dp"
    android:layout_below="@id/button5"
    android:layout_toEndOf="@id/button7"
    android:layout_marginStart="10dp"
    android:layout_marginTop="10dp"
    android:layout_height="60dp"
    android:text="8"
    android:textSize="25dp"
    android:background="#964B00"/>

<Button android:id="@+id/button9"
    android:layout_width="60dp"
    android:layout_below="@id/button6"
    android:layout_toEndOf="@id/button8"
    android:layout_marginStart="10dp"
    android:layout_marginTop="10dp"
    android:layout_height="60dp"
    android:text="9"
    android:textSize="25dp"
    android:background="#964B00"/>

<Button android:id="@+id/buttondot"
    android:layout_width="60dp"
    android:layout_below="@id/button7"
    android:layout_marginStart="50dp"
    android:layout_marginTop="10dp"
    android:layout_height="60dp"
    android:text="."
    android:textSize="25dp"
    android:background="#7FB3D5"/>

<Button android:id="@+id/buttonzero"
    android:layout_width="60dp"
    android:layout_below="@id/button8"
    android:layout_toEndOf="@id/buttondot"
    android:layout_marginStart="10dp"
    android:layout_marginTop="10dp"
    android:layout_height="60dp"
    android:text="0"
    android:textSize="25dp"
    android:background="#7FB3D5"/>

<Button android:id="@+id/buttonC"
    android:layout_width="60dp"
    android:layout_below="@id/button9"
    android:layout_toEndOf="@id/buttonzero"
    android:layout_marginStart="10dp"
    android:layout_marginTop="10dp"
```

```
    android:layout_height="60dp"
    android:text="C"
    android:textSize="25dp"
    android:background="#7FB3D5"/>

  <Button android:id="@+id/buttonadd"
    android:layout_width="90dp"
    android:layout_height="60dp"
    android:layout_toEndOf="@id/button3"
    android:layout_marginStart="10dp"
    android:layout_marginTop="150dp"
    android:text="+"
    android:textSize="25dp"
    android:background="#7FB3D5"/>

  <Button android:id="@+id/buttonsub"
    android:layout_width="90dp"
    android:layout_height="60dp"
    android:layout_below="@id/button1"
    android:layout_toEndOf="@id/button6"
    android:layout_marginStart="10dp"
    android:layout_marginTop="10dp"
    android:text="-"
    android:textSize="25dp"
    android:background="#7FB3D5"/>

  <Button android:id="@+id/buttonmul"
    android:layout_width="90dp"
    android:layout_height="60dp"
    android:layout_below="@id/button4"
    android:layout_toEndOf="@id/button6"
    android:layout_marginStart="10dp"
    android:layout_marginTop="10dp"
    android:text="x"
    android:textSize="25dp"
    android:background="#7FB3D5"/>

  <Button android:id="@+id/buttondiv"
    android:layout_width="90dp"
    android:layout_height="60dp"
    android:layout_below="@id/button7"
    android:layout_toEndOf="@id/button6"
    android:layout_marginStart="10dp"
    android:layout_marginTop="10dp"
    android:text="/"
    android:textSize="25dp"
    android:background="#7FB3D5"/>

  <Button android:id="@+id/buttoneql"
    android:layout_width="300dp"
    android:layout_height="60dp"
    android:layout_below="@id/buttondot"
    android:layout_marginStart="50dp"
    android:layout_marginTop="20dp"
    android:text="="
    android:textSize="25dp"
    android:background="#7FB3D5"/>
</RelativeLayout>
```

SHARED PREFERENCES

Main_activity.java

```
package com.example.sharedpreferences;

import android.content.SharedPreferences;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
   EditText username, mobilenum, email, pass1, pass2;
   Button submit;
   SharedPreferences sharedPreferences;
   SharedPreferences.Editor editor;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      username = findViewById(R.id.username);
      mobilenum = findViewById(R.id.number);
      email = findViewById(R.id.email);
      pass1 = findViewById(R.id.password);
      pass2 = findViewById(R.id.conpassword);
      submit = findViewById(R.id.loginbtn);

      sharedPreferences = getSharedPreferences("UserDetails", MODE_PRIVATE);
      editor = sharedPreferences.edit();

      submit.setOnClickListener(new View.OnClickListener() {
         @Override
         public void onClick(View view) {
            String usernames = username.getText().toString().trim();
            String mobiles = mobilenum.getText().toString().trim();
            String emails = email.getText().toString().trim();
            String pass1s = pass1.getText().toString().trim();
            String pass2s = pass2.getText().toString().trim();

            if (usernames.isEmpty()) {
               username.setError("Username is Empty");
               username.requestFocus();
               return;
            }

            if (mobiles.isEmpty()) {
               mobilenum.setError("Mobile number is Empty");
               mobilenum.requestFocus();
               return;
            }
```

```
            if (emails.isEmpty()) {
                email.setError("Input Email");
                email.requestFocus();
                return;
            }

            if (pass1s.isEmpty()) {
                pass1.setError("Enter Password");
                pass1.requestFocus();
                return;
            }

            if (pass1s.length() < 6) {
                pass1.setError("Length must be minimum 6 characters");
                pass1.requestFocus();
                return;
            }

            if (!pass1s.equals(pass2s)) {
                pass2.setError("Password not Matched");
                pass2.requestFocus();
                return;
            }
            Toast.makeText(MainActivity.this, "Registration Successful", Toast.LENGTH_LONG).show();

            editor.putString("keyusername", usernames);
            editor.putString("keymobile", mobiles);
            editor.putString("keyemail", emails);
            editor.putString("keypassword", pass2s);
            editor.apply();
        }
    });
  }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity">

  <TextView
    android:id="@+id/signin"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginStart="50dp"
    android:layout_marginEnd="50dp"
    android:layout_marginTop="50dp"
    android:layout_marginBottom="50dp"
```

```
    android:gravity="center"
    android:text="MES COLLEGE OF ENGINEERING"
    android:background="@color/design_default_color_primary_variant"
    android:textColor="@color/white"
    android:textSize="30dp"
    android:textStyle="bold" />

  <EditText
    android:id="@+id/username"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@+id/signin"
    android:layout_marginStart="20dp"
    android:layout_marginTop="10dp"
    android:layout_marginEnd="20dp"
    android:background="#eeccff"
    android:drawableLeft="@drawable/baseline_account_circle_24"
    android:drawablePadding="20dp"
    android:hint="Username"
    android:padding="20dp" />

  <EditText
    android:id="@+id/number"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@+id/username"
    android:layout_marginStart="20dp"
    android:layout_marginTop="20dp"
    android:layout_marginEnd="20dp"
    android:background="#eeccff"
    android:drawableLeft="@drawable/baseline_add_call_24"
    android:drawablePadding="20dp"
    android:hint="Mobile Number"
    android:padding="20dp" />

  <EditText
    android:id="@+id/email"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@+id/number"
    android:layout_marginStart="20dp"
    android:layout_marginTop="26dp"
    android:layout_marginEnd="20dp"
    android:background="#eeccff"
    android:drawableLeft="@drawable/baseline_contact_mail_24"
    android:drawablePadding="20dp"
    android:hint="Email"
    android:padding="20dp" />

  <EditText
    android:id="@+id/password"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@+id/email"
    android:layout_marginStart="20dp"
    android:layout_marginTop="20dp"
    android:layout_marginEnd="20dp"
    android:background="#eeccff"
```

```
    android:drawableLeft="@drawable/baseline_gpp_good_24"
    android:drawablePadding="20dp"
    android:hint="Password"
    android:inputType="textPassword"
    android:padding="20dp" />

  <EditText
    android:id="@+id/conpassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@+id/password"
    android:layout_marginStart="20dp"
    android:layout_marginTop="20dp"
    android:layout_marginEnd="28dp"
    android:layout_marginBottom="26dp"
    android:background="#eeccff"
    android:drawableLeft="@drawable/baseline_gpp_good_24"
    android:drawablePadding="20dp"
    android:hint="Confirm Password"
    android:inputType="textPassword"
    android:padding="26dp" />

  <com.google.android.material.button.MaterialButton
    android:id="@+id/loginbtn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/conpassword"
    android:layout_centerHorizontal="true"
    android:layout_marginStart="10dp"
    android:text="SUBMIT" />

</RelativeLayout>
```

EXPLICIT INTENT

MAIN_ACTIVITY.JAVA

```
package com.example.explicitintent;

import androidx.appcompat.app.AppCompatActivity;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
   }

   public void onIndexSendButtonClicked(View view) {

      EditText editText = findViewById(R.id.text);
      String index = editText.getText().toString().trim();

      if (index.isEmpty()) {
         Toast.makeText(
               this,
               "Please enter an image index",
               Toast.LENGTH_SHORT
         ).show();
         return;
      }

      Intent intent = new Intent(
            MainActivity.this,
            MainActivity2.class
      );

      intent.putExtra("ImageIndex", index);
      startActivity(intent);
   }
}
```

Main_activity2.java

```
package com.example.explicitintent;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.ImageView;
import android.widget.Toast;
```

```
public class MainActivity2 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);

        String index = getIntent().getStringExtra("ImageIndex");

        if (index != null) {
            setImage(index);
        } else {
            Toast.makeText(
                    this,
                    "No image index received",
                    Toast.LENGTH_SHORT
            ).show();
            finish();
        }
    }

    private void setImage(String index) {

        ImageView imageView = findViewById(R.id.img1);

        switch (index) {
            case "1":
                imageView.setImageResource(R.drawable.apple);
                break;

            case "2":
                imageView.setImageResource(R.drawable.pineapple);
                break;

            case "3":
                imageView.setImageResource(R.drawable.orange);
                break;

            case "4":
                imageView.setImageResource(R.drawable.pomegranate);
                break;

            default:
                Toast.makeText(
                        this,
                        "Invalid image index. Enter 1 to 4.",
                        Toast.LENGTH_SHORT
                ).show();
                finish();
                break;
        }
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="16dp"
   tools:context=".MainActivity">

   <EditText
      android:id="@+id/text"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="@string/write_image_index"
      android:inputType="number"
      android:singleLine="true" />

   <Button
      android:id="@+id/btn"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="@string/send_image_index"
      android:onClick="onIndexSendButtonClicked" />

</LinearLayout>
```

Activity_main2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="16dp"
   tools:context=".MainActivity2">

   <ImageView
      android:id="@+id/img1"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:scaleType="fitCenter"
      android:contentDescription="Selected image" />

</LinearLayout>
```